# The coefficient at the penultimate Markov-sail vertex

A proof of MathDB 366808
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**Task 1.** Prove the Location of 4 conjecture of Evans, Veselov, and Winn [1, Conjecture 8.4]. We use their numerator convention

$$M_{a/b}(x,y,z)=\frac{P_{a/b}(x^2,y^2,z^2)}{x^{a-1}y^{b-1}z^{a+b-1}},\qquad P_{a/b}(u,v,w)=\sum_{i,j}A_{ij}u^iv^jw^{a+b-1-i-j}.$$

For coprime $2\le a<b$, the conjecture specifies a coefficient in the interior of the critical triangle $i<a$, $j<b$, $bi+aj>ab$. If $b/a=[h_1,\dots,h_n]$, with convergents $p_k/q_k$, its sail coordinates are

$$A_m^{\mathrm{sail}}=(q_{2m-1},b-p_{2m-1}),\qquad B_m^{\mathrm{sail}}=(a-q_{2m},p_{2m}).$$

The claimed coefficient is 4 at $A_m^{\mathrm{sail}}$ when $n=2m$, and at $B_m^{\mathrm{sail}}$ when $n=2m+1$. The source defines sail coefficients only in that interior; for $a=1$ the critical triangle is empty, as it explicitly notes in Section 9.

**Theorem 1.** *For every coprime pair $2\le a<b$, there is a unique pair of integers $0<i<a$, $0<j<b$ satisfying*

$$bi+aj=ab+1.$$

*The coefficient $A_{ij}$ of $P_{a/b}$ at this pair is 4. Consequently Conjecture 8.4 holds in both parity cases.*

*Proof.* We first recall two established facts from [1, Theorems 3.2 and 4.2]. For coprime positive $r,s$, the Newton polygon of $P_{r/s}(u,v,1)$ is

$$\left\{(i,j):i,j\ge 0,\quad \frac{i}{r}+\frac{j}{s}\ge 1,\quad i+j\le r+s-1\right\}.$$

Its coefficients at $(r,0)$ and $(0,s)$ are both 1. No assertion of saturation of the polygon is needed.

Work in formal Laurent series in $t$, and set

$$\kappa(t)=\frac{1+t^{2a}+t^{2b}}{t^{a+b}},\qquad F_{r/s}(t)=\kappa(t)M_{r/s}(t^b,t^a,1).$$

The numerator convention gives the useful identity

$$F_{r/s}(t)=(1+t^{2a}+t^{2b})\,t^{-(br+as)}P_{r/s}(t^{2b},t^{2a},1). \tag{1}$$

If $|br-as|=1$, the linear weight $bi+aj$ has a unique minimum $\min(br,as)$ on the Newton polygon, at one of its two lower endpoints. The minimum coefficient is 1. All weight differences are integers, so (1) implies

$$F_{r/s}(t)=t^{-1}+O(t). \tag{2}$$

Here and below the $O$ notation denotes divisibility in the ring of formal Laurent series.

Let $r/s$ and $u/v$ be the Farey parents of $a/b$, so that $r+u=a$, $s+v=b$, and $rv-su=\pm1$. Both parents have positive numerator and denominator since $a\ge 2$. Then $|br-as|=|bu-av|=1$. The three associated Markov polynomials form a triple satisfying the generalized Markov equation. After multiplication by $\kappa$, the series

$$U=F_{r/s},\qquad V=F_{u/v},\qquad W=F_{a/b}$$

therefore satisfy

$$U^2 + V^2 + W^2 = UVW, \qquad UV(W-2) = (U-V)^2 + W^2. \tag{3}$$

For the target polygon, the minimum weight is $ab$ along the segment from $(a,0)$ to $(0,b)$. Coprimality implies that its only lattice points are its endpoints, each with coefficient 1. Thus $W = 2 + O(t^2)$. By (2), $U - V = O(t)$ and $UV = t^{-2}(1 + O(t^2))$. Substituting these facts into (3) yields

$$W - 2 = \frac{(U-V)^2 + W^2}{UV} = 4t^2 + O(t^4). \tag{4}$$

Because $a, b \geq 2$, the factor $1 + t^{2a} + t^{2b}$ in (1) does not change the coefficient of $t^2$ in $W$. That coefficient is therefore

$$\sum_{bi+aj=ab+1} A_{ij} = 4. \tag{5}$$

There is exactly one nonnegative integer solution to $bi + aj = ab + 1$. Indeed choose the unique $i \in \{1, \ldots, a-1\}$ with $bi \equiv 1 \pmod{a}$ and put $j = (ab + 1 - bi)/a$. Then $0 < j < b$, and any other integer solution differs by $(ka, -kb)$. Neither $k > 0$ nor $k < 0$ preserves nonnegativity. This point also satisfies $i + j \leq a + b - 2$, so it lies in the critical triangle and in the stated Newton polygon. Equation (5) proves $A_{ij} = 4$.

Finally put $(p, q) = (p_{n-1}, q_{n-1})$ for the penultimate convergent. With the source's indexing $p_0 = 1, q_0 = 0$, the continued-fraction determinant identity gives

$$bq - ap = (-1)^n.$$

If $n$ is even, the point $(q, b-p) = A_m^{\text{sail}}$ has weight $bq + a(b-p) = ab + 1$. If $n$ is odd, the point $(a-q, p) = B_m^{\text{sail}}$ has weight $b(a-q) + ap = ab + 1$. Both are interior points, and uniqueness identifies each with the point in (5). This proves both assertions of the conjecture. □

**Task 2.** Determine explicit formulas for the coefficients at lattice heights $bi + aj - ab = 2, 3, \ldots$ by expanding the same formal Markov identity to higher orders.

**Task 3.** Prove the Factor 4 conjecture for every coefficient in the critical triangle, extending the height-one result beyond the penultimate vertex.

**Task 4.** Use valuations adapted to other sail vertices to prove the Markov Sail Duality identities for arithmetic progressions of coefficients along sail segments.

**Task 5.** Give a direct perfect-matching interpretation of the four objects counted by the coefficient at the penultimate sail vertex.

## References


[1] S. J. Evans, A. P. Veselov, and B. Winn, *Arithmetic and geometry of Markov polynomials*, arXiv:2501.14882v3 (2025); current 28-page journal-hosted manuscript, Theorems 3.2 and 4.2, and Conjecture 8.4.
`https://armj.math.stonybrook.edu/Articles/250708-Winn/article.pdf`.